

Multi-task Medical Image Analysis with Diffusion-Based Synthetic Augmentation

Clinical Performance Report

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Dataset	Breast Ultrasound Images (BUSI)	Report Date	2026-06-13 14:40

1. Segmentation Results (U-Net)

Metric	Baseline (Real Only)	Augmented (Best Ratio)	Target
Dice Score	0.5469	0.5469	> 0.75
IoU	0.4707	0.4707	> 0.60
Sensitivity	0.5861	0.5861	> 0.80
Specificity	0.9785	0.9785	> 0.85

2. Classification Results (ResNet-50)

Metric	Baseline (Real Only)	Augmented (Best Ratio)	Target
AUC (OvR)	0.9996	0.9126	> 0.85
Accuracy	0.9833	0.8050	> 0.80
F1 – Normal	0.9744	—	> 0.78
F1 – Benign	0.9857	—	> 0.78
F1 – Malignant	0.9836	—	> 0.78

3. Synthetic Image Quality (Diffusion Model)

Metric	Value	Interpretation
FID Score	218.7900	Lower is better (real ↔ synthetic similarity)
Images Generated	450	Total synthetic images across all classes

4. Ablation Study — Synthetic Data Ratio vs Performance

Synthetic Ratio	Val AUC	Val Accuracy	Improvement vs Baseline
0%	0.9666	0.9000	—
25%	0.9126	0.8050	-0.0540
50%	0.9042	0.7875	-0.0624
75%	0.8965	0.7440	-0.0701
100%	0.8769	0.7720	-0.0897

